Cloud CI Lab Project - Submission Report

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1. GitHub Repository Link

https://github.com/eshunvera62-source/cloud-ci-lab-project

2. ci.yml File

```
name: Cloud CI Pipeline
on:
  push:
    branches: ["main"]
  pull_request:
    branches: ["main"]
  workflow_dispatch:

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.11"

      - name: Print runtime version
```

run: python --version

- name: Echo build message

  run: echo "Starting Cloud CI Pipeline build..."

- name: Run app

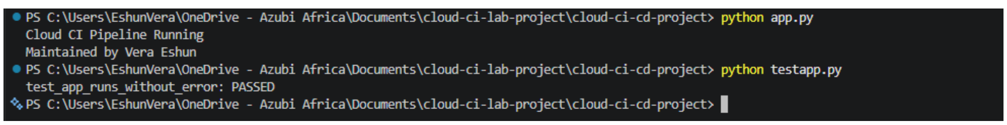
  run: python app.py

- name: Run tests

  run: python test_app.py

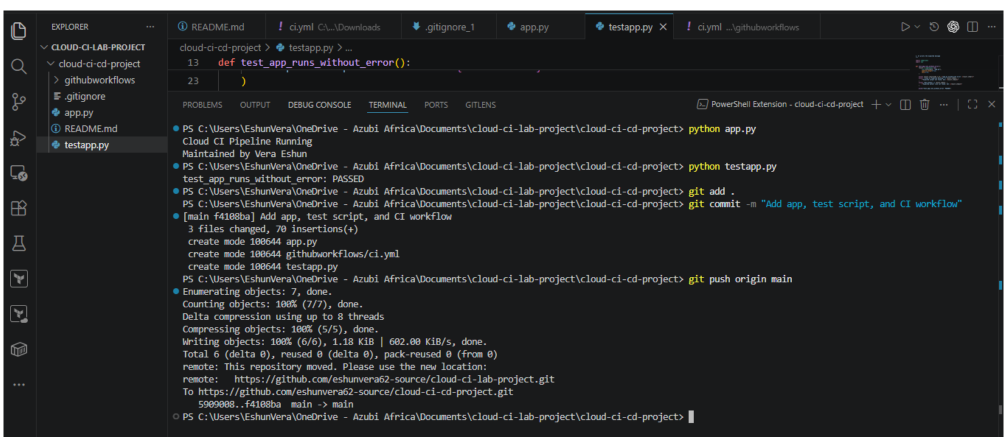
3. Screenshots

Screenshot 1: Repository URL



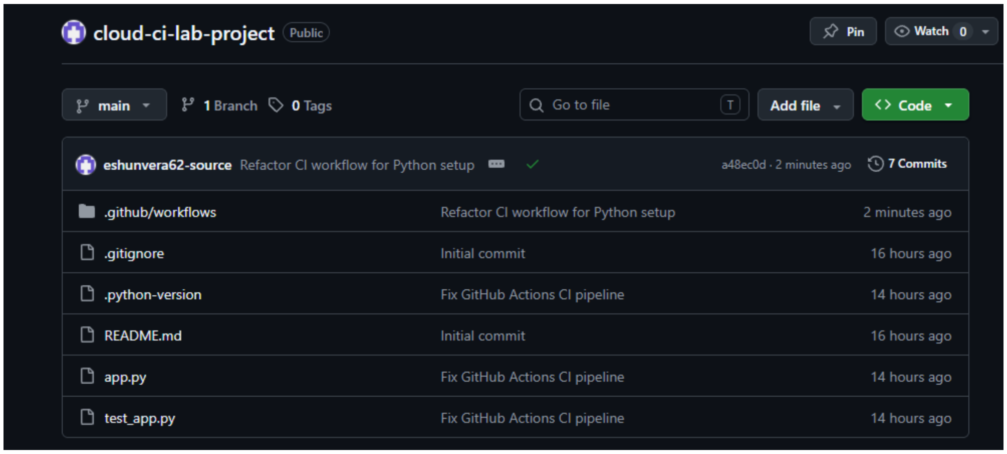
Note: Shows the URL of the GitHub repository created for this project (Task 1).

Screenshot 2: GitHub Actions Setup Page



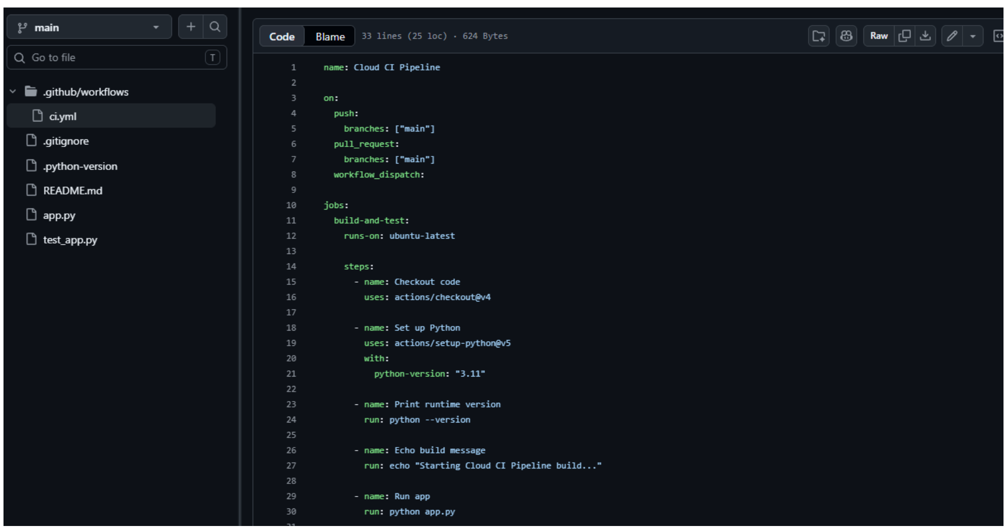
Note: The GitHub Actions tab before the workflow file was added, showing the option to set up a workflow (Task 4).

Screenshot 3: Workflow File Editor



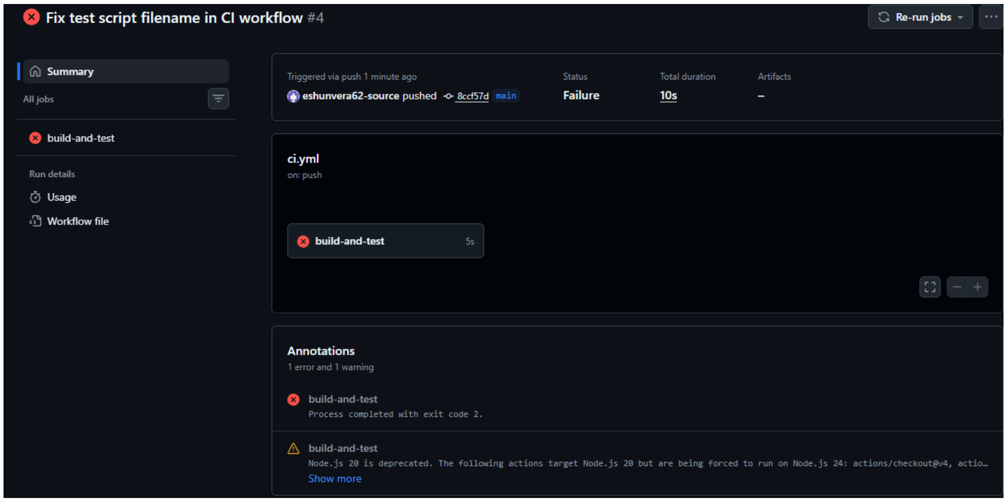
Note: The ci.yml file being created and edited directly in the GitHub web editor (Task 4).

Screenshot 4: Failed Pipeline Run



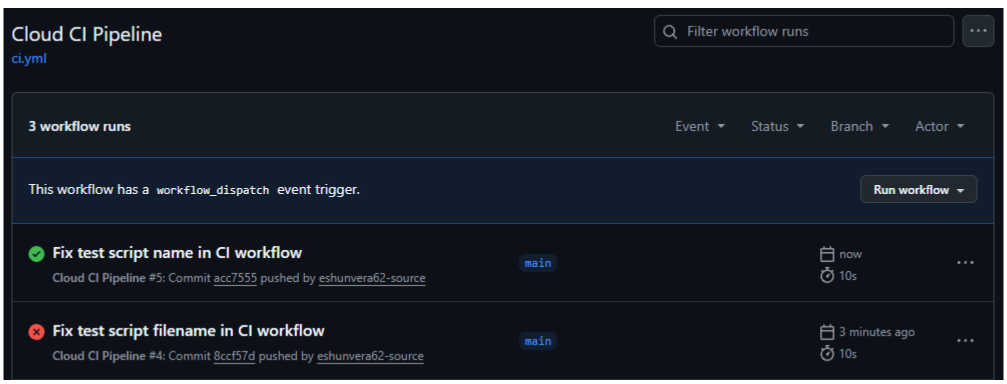
Note: A pipeline run that failed after an intentional error was introduced, showing the Failure status (Task 6).

Screenshot 5: Error Identified



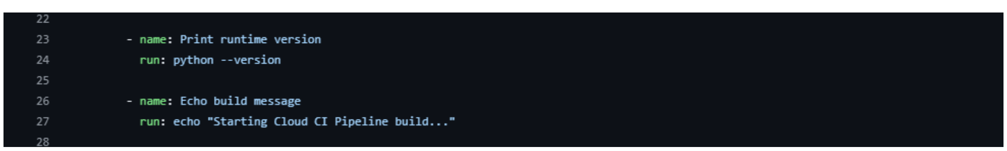
Note: The workflow error message that was identified and corrected during troubleshooting (Task 6).

Screenshot 6: Fixed and Passing Run



Note: The repository showing the pipeline passing with a green checkmark after the error was fixed (Task 6).

Screenshot 7: Extended Pipeline Confirmed



Note: Confirmation that the pipeline, including the added runtime version and build message steps, completed successfully (Task 7).

4. Short Note: What did your pipeline automate?

This pipeline automates the build and validation of a simple Python application. On every push to the main branch, GitHub Actions automatically spins up a fresh Ubuntu environment, checks out the latest code, installs Python 3.11, prints the runtime version, echoes a build start message, runs the application to confirm it executes without error, and then runs the test script to validate its output. This removes the need for manual testing after every change and immediately flags any broken code through a failed run, as demonstrated when an intentional error was introduced and caught by the pipeline before being fixed.